

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("SampleSuperstore.csv", encoding='latin1')
```

```
print(df.head())
print(df.info())
print(df.columns)
```

```
4 Sean O'Donnell Consumer United States Fort Lauderdale ...
```

	Postal Code	Region	Product ID	Category	Sub-Category	\
0	42420	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311	South	FUR-TA-10000577	Furniture	Tables	
4	33311	South	OFF-ST-10000760	Office Supplies	Storage	

	Product Name	Sales	Quantity	\
0	Bush Somerset Collection Bookcase	261.9600	2	
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2	Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4	Eldon Fold 'N Roll Cart System	22.3680	2	

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164

```
[5 rows x 21 columns]
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 9994 entries, 0 to 9993
```

```
Data columns (total 21 columns):
```

#	Column	Non-Null Count	Dtype
0	Row ID	9994 non-null	int64
1	Order ID	9994 non-null	object
2	Order Date	9994 non-null	object
3	Ship Date	9994 non-null	object
4	Ship Mode	9994 non-null	object
5	Customer ID	9994 non-null	object
6	Customer Name	9994 non-null	object
7	Segment	9994 non-null	object
8	Country	9994 non-null	object
9	City	9994 non-null	object
10	State	9994 non-null	object
11	Postal Code	9994 non-null	int64
12	Region	9994 non-null	object
13	Product ID	9994 non-null	object
14	Category	9994 non-null	object
15	Sub-Category	9994 non-null	object
16	Product Name	9994 non-null	object
17	Sales	9994 non-null	float64
18	Quantity	9994 non-null	int64
19	Discount	9994 non-null	float64
20	Profit	9994 non-null	float64

```
dtypes: float64(3), int64(3), object(15)
```

```
memory usage: 1.6+ MB
```

```
None
```

```
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
      'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
      'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
      'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
      dtype='object')
```

```
print(df.isnull().sum())
```

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0

```

Segment      0
Country      0
City         0
State        0
Postal Code  0
Region       0
Product ID   0
Category     0
Sub-Category 0
Product Name 0
Sales        0
Quantity     0
Discount     0
Profit       0
dtype: int64

```

```
df.drop_duplicates(inplace=True)
```

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
```

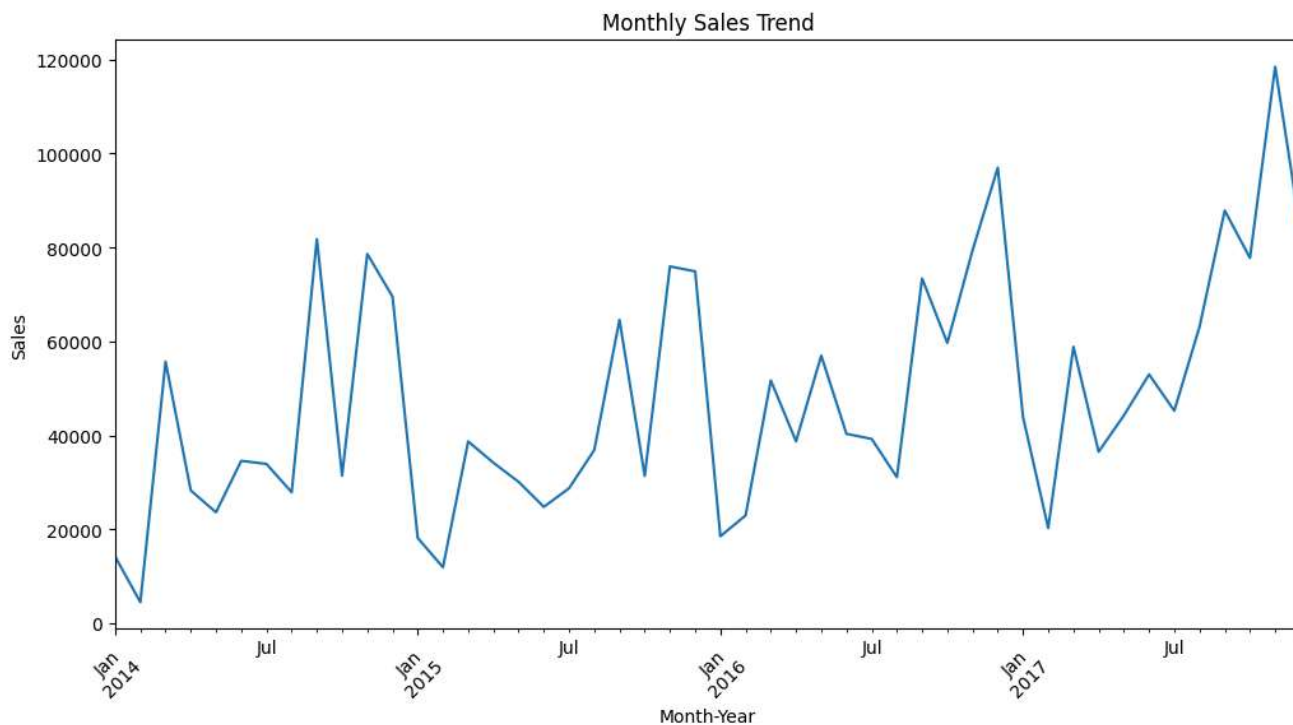
```
df['Month_Year'] = df['Order Date'].dt.to_period('M')
```

```
monthly_sales = df.groupby('Month_Year')['Sales'].sum()
```

```

plt.figure(figsize=(12,6))
monthly_sales.plot()
plt.xticks(rotation=45)
plt.title("Monthly Sales Trend")
plt.xlabel("Month-Year")
plt.ylabel("Sales")
plt.show()

```

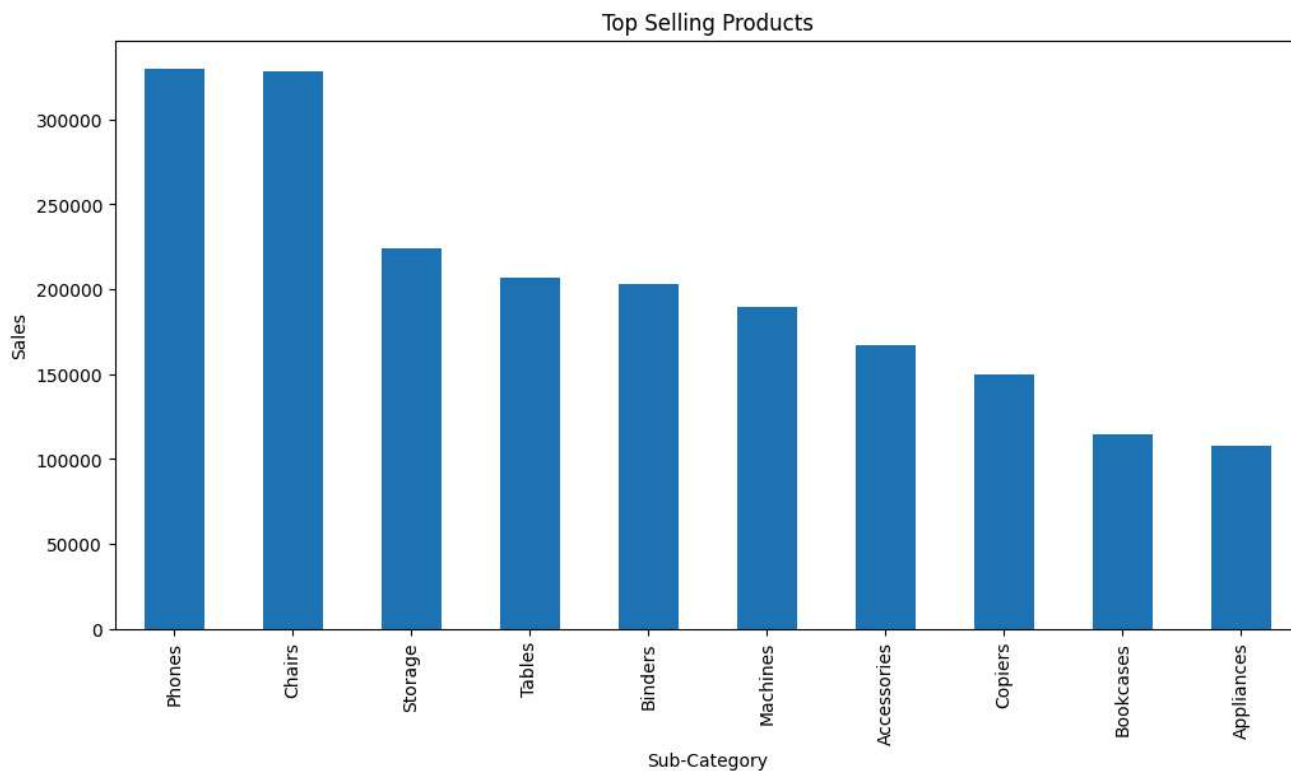


```
top_products = df.groupby('Sub-Category')['Sales'].sum().sort_values(ascending=False)
```

```

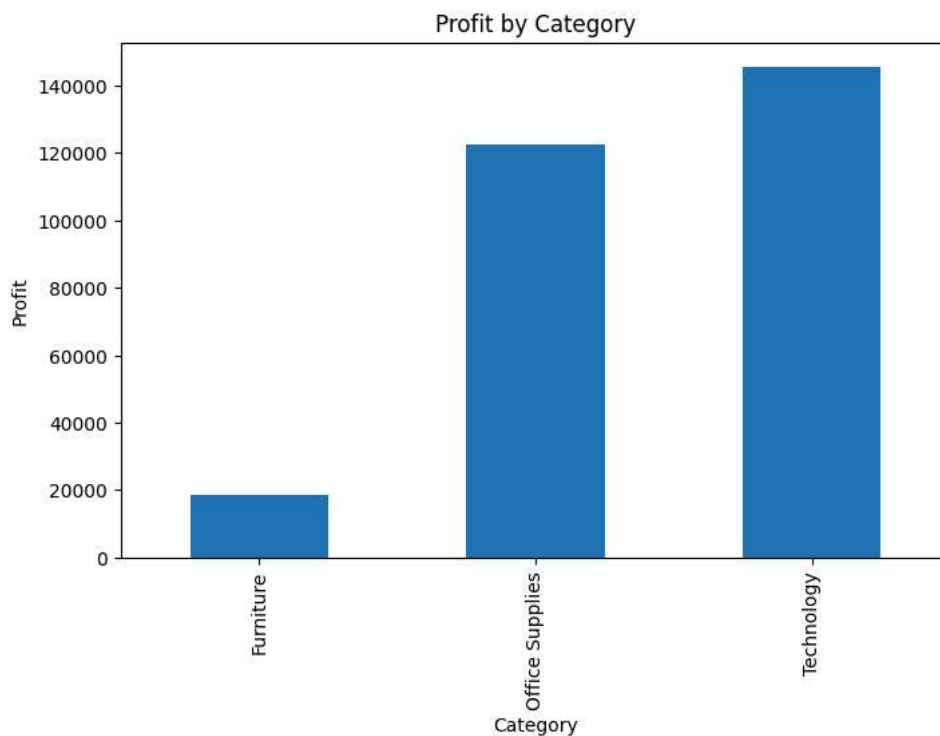
plt.figure(figsize=(12,6))
top_products.head(10).plot(kind='bar')
plt.title("Top Selling Products")
plt.xlabel("Sub-Category")
plt.ylabel("Sales")
plt.show()

```



```
profit_category = df.groupby('Category')['Profit'].sum()
```

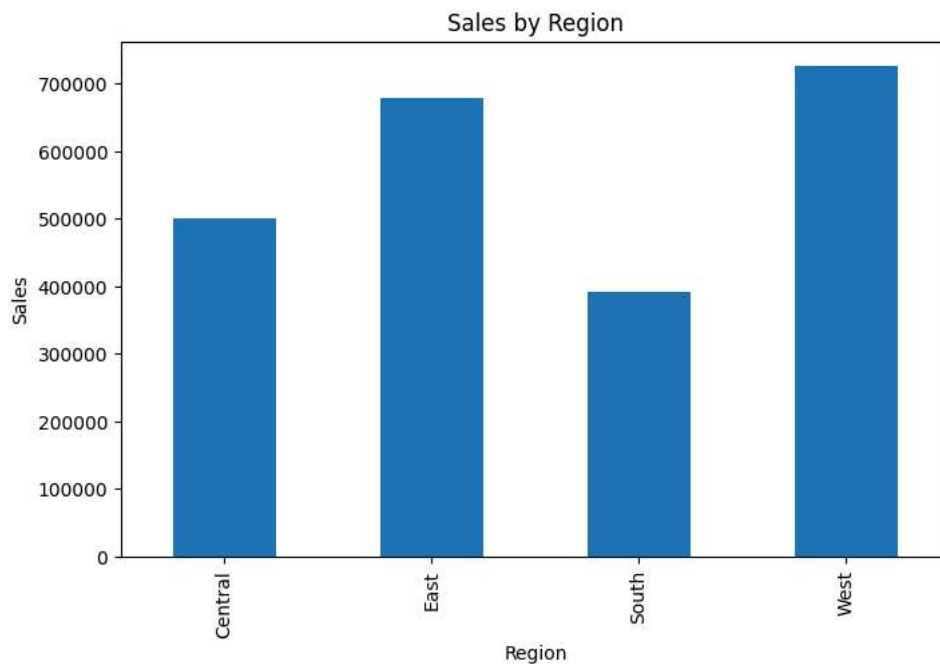
```
plt.figure(figsize=(8,5))  
profit_category.plot(kind='bar')  
plt.title("Profit by Category")  
plt.ylabel("Profit")  
plt.show()
```



```
region_sales = df.groupby('Region')['Sales'].sum()
```

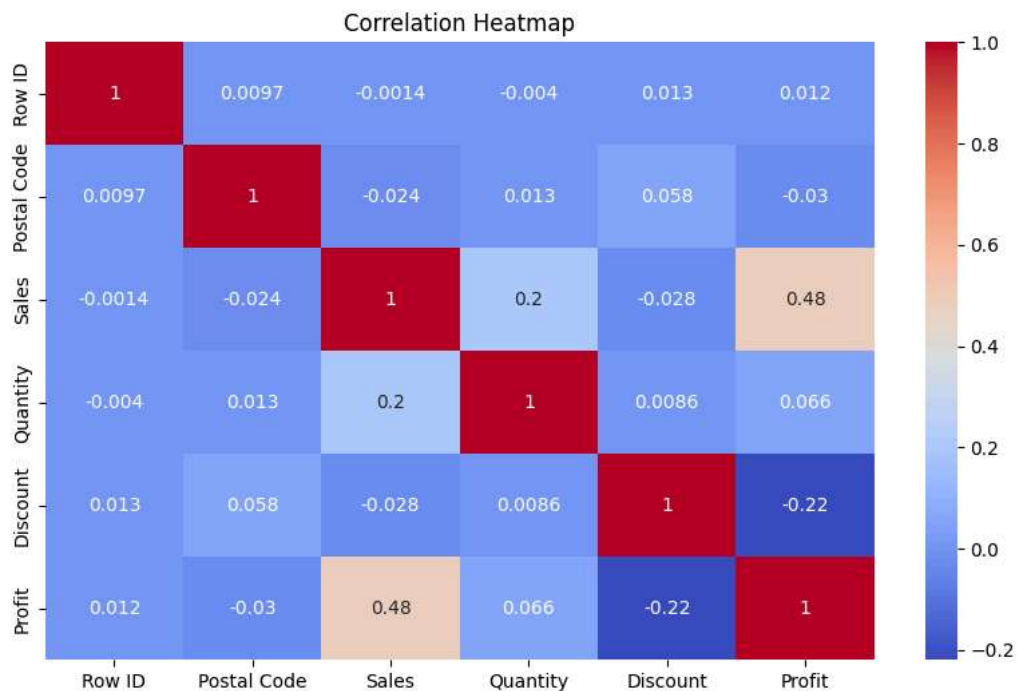
```
plt.figure(figsize=(8,5))
```

```
region_sales.plot(kind='bar')
plt.title("Sales by Region")
plt.ylabel("Sales")
plt.show()
```



```
numeric_df = df.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



1. Technology category generated highest sales.
2. Western region contributed maximum revenue.

3. Some categories had high sales but low profit.
4. Sales increased during specific months indicating seasonal trends.

The analysis identified sales trends, profitable categories, and regional performance patterns. These insights can help businesses optimize inventory and marketing strategies.

```
df.to_csv("cleaned_superstore.csv", index=False)
```